

Lab 4

Attack Analysis

Cameron Bergeron

12/5/2025

Table of Contents

Introduction.....	1
Investigation Methodology.....	1
1. Wireshark.....	1
1.1 - DNS Inspection.....	2
1.2 - SMB Analysis.....	2
2. Event Viewer.....	2
Findings.....	2
1. Identification of Systems.....	2
2. Reconnaissance Activity.....	3
3. SMB Exploitation Behavior.....	3
4. Host Logging Gaps and Silent Exploitation.....	4
5. Interpretation of Findings.....	4
Attack Timeline.....	5
Future Recommendations & Conclusion.....	7
References.....	8
Appendix.....	9
• Figure 1: Screenshot of the Capture.pcapng Properties.....	9
• Figure 2: Screenshot of the Capture.pcapng Protocol Hierarchy.....	9
• Figure 3: Screenshot of the DNS Filter.....	10
• Figure 4: Screenshot of the SMB Filter.....	10
• Figure 5: Screenshot of the Frames Containing MZ.....	11
• Figure 6: Screenshot of Event Viewer.....	11
• Figure 7: Screenshot of the Network Endpoints.....	12
• Figure 8: Screenshot of the SYN flag Filter.....	12

Introduction

This report documents a complete forensic attack analysis conducted on a potentially hacked system. The CISO declared an incident following suspicious activity, and a Cyber First Responder collected preliminary forensic evidence before leaving the case. As the assigned Security Analyst, my task was to analyze the provided network packet captures and Windows event logs. My goal was to determine the nature of the attack, identify the compromised system, reconstruct the attack timeline, and provide mitigation recommendations.

Using Wireshark and Window's Event Viewer, I analysed the following artifacts:

- Capture.pcapng – full network packet capture during the observed compromise
- Maybe Hacked Box – Security Log.evt
- Maybe Hacked Box – Application Log.evt

The key goals of this investigation were to:

1. Identify the attacker and victim systems
2. Determine how the attacker gained access
3. Identify malware or payloads delivered
4. Reconstruct the sequence of events
5. Recommend corrective and preventative actions

Investigation Methodology

1. Wireshark

To begin investigating, I used Wireshark to analyze the Capture.pcapng file. See [Figure 1](#) for a screenshot of the captured file properties. To gain a better understanding of what needed to be investigated, I examined the protocol hierarchy of the captured file. See [Figure 2](#) for a screenshot of the protocol hierarchy. Based on what I saw, wireshark was used to analyze traffic flows, detect anomalies, identify potential exploit behavior, and examine SMB transactions.

- Filters used included:
 - tcp.flags.syn == 1 && tcp.flags.ack == 0 (scan detection)
 - smb (SMB activity)
 - frame contains "MZ" (malware file transfer detection)
 - Protocol hierarchy and endpoint statistics for context

1.1 - DNS Inspection

Initial DNS inspection revealed high-volume queries to Microsoft NCSI and Teredo IPv6 services. See [Figure 3](#) for a screenshot of the DNS filter. While these domains are legitimate, the frequency and persistence may indicate attempts by malware to exploit IPv6 tunneling or hide outbound communications.

1.2 - SMB Analysis

See [Figure 4](#) for a screenshot of the SMB filter. SMB Write operations from 192.168.134.132 to 192.168.134.129 show possible file transfer or malware propagation. Data size patterns indicate small-to-medium payloads consistent with executable or script staging.

During SMB analysis, I filtered for *frame contains "MZ"* and observed PE header bytes (MZ) embedded within SMB traffic from 192.168.134.129 to 192.168.134.132. See [Figure 5](#) for a screenshot of the frames containing MZ. This indicates that an executable file was being transferred over SMB, most likely as part of malware propagation or remote code execution.

2. Event Viewer

To analyze Security Log.evt and Application Log.evt, I used Window's Event Viewer. See [Figure 6](#) for a screenshot of Event Viewer. Review of the Security Log revealed only service start/stop events (7035/7036), EventLog startup/shutdown artifacts (6005/6006/6009), and miscellaneous system messages. No authentication events (4624/4625), no privilege escalation events (4672), no process creation events (4688), and no service installation events (7045) were present in the Security Log.

This absence of expected security auditing strongly suggests that the attacker did not authenticate normally. Instead, the attacker likely used a remote SMB exploit such as MS08-067 or MS17-010. Such exploits allow remote code execution over named pipes (\IPC\$, \SPOOLSS, \SRVSVC) without generating logon entries or creating normal process audit trails.

The complete lack of host-level audit events, combined with Wireshark evidence of SMB-based payload delivery (MZ executable segments), internal scanning, and named-pipe activity confirms that the attacker bypassed standard Windows authentication and logging entirely.

Findings

1. Identification of Systems

Using Wireshark to analyze the network endpoints revealed:

- 192.168.134.129 - Attacker System
 - Heavy scanning activity, SMB exploitation, malware transfer.
- 192.168.134.132 - Victim System
 - Target of SMB exploitation and payload delivery.

Other hosts (e.g., 192.168.134.158) showed typical benign Windows DNS traffic. See [Figure 7](#) for a screenshot of the network endpoints.

2. Reconnaissance Activity

Using the SYN flag filter:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

192.168.134.129 (attacker system) was observed performing a high-volume port scan against 192.168.134.132 (victim system), hitting ports including:

- 445 (SMB)
- 3389 (RDP)
- 80/443 (HTTP/HTTPS)
- 161 (SNMP)
- 25 (SMTP)

This is characteristic of internal reconnaissance or worm-like lateral movement. See [Figure 8](#) for a screenshot of the SYN flag filter.

3. SMB Exploitation Behavior

The most critical forensic evidence comes from the SMB traffic:

SMB sessions showed high activity involving pipes such as:

- \IPC\$
- \SPOOLSS
- \SRVSVC
- \PIPE\browser

These pipes are commonly abused by SMB remote code execution vulnerabilities, especially:

- MS08-067 (Server Service exploit)
- MS17-010 (EternalBlue family exploit)

Such exploits grant attackers remote execution without authentication, which becomes important when correlating logs. Additionally, large SMB write operations were found. There were repeated “Write AndX” operations transferring large blocks of binary data to the victim.

Furthermore, when filtering for *frame contains "MZ"* it revealed Windows Portable Executable (PE) header signatures embedded inside SMB traffic. This shows conclusively that an executable malware file was transferred over SMB. This most likely represents malware staging or worm propagation.

4. Host Logging Gaps and Silent Exploitation

The Windows Security Log showed none of the following expected events:

- 4624 Successful Login
- 4625 Failed Login
- 4672 Admin Privilege Assigned
- 4688 Process Creation
- 7045 Service Installation

Instead, logs contained only normal system service activity (7035/7036), EventLog startup messages, driver initialization, and benign system events.

- However, this absence is itself forensic evidence:
 - The attacker did not authenticate using valid credentials.
 - The attacker did not trigger process creation logs.
 - The compromise bypassed Windows security auditing entirely.

This behavior is typical of SMB remote exploit attacks, which execute code via vulnerable SMB services without generating host-level logs. Thus, the logs confirm exploitation rather than credential compromise.

5. Interpretation of Findings

Analysis of the packet capture revealed clear signs of malicious SMB behavior, including large volumes of SMB Write operations and the transfer of an executable file identified by its MZ header signature. The attacking host (192.168.134.129) first

performed a wide port scan against 192.168.134.132, then initiated SMB communication using named pipes such as IPC\$, \SRVSVC, and \SPOOLSS, which are commonly abused by known vulnerabilities including MS08-067 and MS17-010.

Notably, the corresponding Windows Security Logs showed no successful or failed logon events, no service installation events, and no process creation events, strongly suggesting that the attacker did not authenticate normally but instead leveraged a remote execution exploit that bypassed Windows authentication and auditing. This is consistent with exploit-based lateral movement where attackers deliver and execute malicious payloads directly over SMB without generating host-level login activities.

Attack Timeline

Timestamp	Source → Destination	Event / Evidence	Technical Interpretation	Kill Chain Phase
0–5 sec	System Boot Logs	EventLog 6005, 6009 appear, indicating the Windows XP/2000-era system has recently booted.	Confirms system was in startup state; no signs of user logon.	<i>Preparation / Baseline</i>
~7.9 sec	192.168.134.129 → 192.168.134.132	First SMB session establishment packets; named pipes (IPC\$, Browser, SPOOLSS) queried.	Host 129 begins probing SMB services on 132.	Reconnaissance
7–10 sec	192.168.134.129 → 192.168.134.132	SMB dialect negotiation with NTLMSSP data; no associated Windows logon event (4624).	Indicates SMB session created without authentication → suggests SMB exploit.	Initial Access (Exploit)

7–40 sec	192.168.134.129 → 192.168.134.132	SMB write operations detected, including named pipes: \SRVSVC, \SPOOLSS, \PIPE\browser.	These pipes are often used in MS08-067 / MS17-010 exploitation for RCE.	Exploitation / Remote Code Execution
~8–12 sec	192.168.134.129 → 192.168.134.132	Continuous SYN packets to ports: 80, 443, 445, 3389, 161, 25, etc. (tcp.flags.syn==1 && tcp.flags.ack==0)	Classic internal port scan, enumerating reachable services.	Reconnaissance
10–40 sec	192.168.134.129 → 192.168.134.132	Increasing volume of SMB Write AndX commands—several hundred writes.	Attacker is sending staged payloads or copying malware over SMB.	Lateral Movement
~25–40 sec	192.168.134.129 → 192.168.134.132	frame contains "MZ" flags executable file data transferred inside SMB stream.	Confirms transfer of a Windows PE malware executable.	Malware Delivery
Throughout capture	192.168.134.158 → DNS server	Repeated DNS queries to dns.msftncsi.com, teredo.ipv6.microsoft.com.	Normal Windows behavior, not tied to attack; used for baseline analysis.	<i>Benign Noise</i>
No corresponding security events	—	No 4624, 4625, 4672, 4688, 7045 appear in any Security Log entries.	Confirms attacker did not authenticate, did not create processes via normal OS	Stealth / Anti-Forensics

			mechanisms, indicating exploit-based access.	
SMB Named-Pipe Activity	192.168.134.129 ↔ 192.168.134.132	Pipes such as \SPOOLSS, \SRVSVC, and \BROWSER heavily used.	These pipes map directly to SMB RCE vulnerabilities used by worms (Conficker, WannaCry family) and manual exploitation tools.	Post-Exploitation Control Channel
End of capture (~40 sec)	192.168.134.129 → 192.168.134.132	Large sustained SMB traffic continues; payload appears partially transferred.	Indicates ongoing compromise; likely next phase would be remote execution of delivered malware.	Persistence / Execution (attempted)

Future Recommendations & Conclusion

The attacker scanned the internal network, discovered the SMB service running on the victim host, and leveraged a remote code execution vulnerability (likely MS08-067 or MS17-010) to gain access without authentication. The attacker then delivered an executable malware payload via SMB Write operations. Because the attack used an exploit rather than a credentialed login, the host Security Log contains no trace of authentication or process logs, confirming silent exploitation. This is a classic lateral movement and worm-propagation pattern seen in early Windows-based SMB attacks. The potential impact of such an attack may include malware installation, remote code execution, potential ransomware staging, or loss of confidentiality within the local network segment.

Going forward, there are some immediate actions needed to handle and prevent this situation:

1. Isolate affected hosts (192.168.134.129 and 192.168.134.132).

2. Image affected systems for deeper forensic analysis.
3. Block SMBv1 at the firewall and disable SMBv1 on all hosts.
4. Apply patches for MS08-067 and MS17-010 (if still applicable depending on OS version).
5. Conduct malware scans and verify system integrity.

Additionally, there are steps needed to long-term harden your system/network:

1. Enable full Windows Security Auditing (Logon/logoff, Process creation, Object access)
2. Deploy network segmentation to limit lateral movement paths.
3. Implement IDS/IPS signatures for SMB exploitation detection.
4. Upgrade legacy operating systems.
5. Enforce patch management policies ensuring timely updates.

Ultimately, this investigation highlights how quickly an unpatched and unmonitored system can be compromised through silent, exploit-based lateral movement. Although the attacker left no trace in the Windows security logs, the network evidence provided a clear and compelling picture of reconnaissance, exploitation, and malware delivery over SMB. This incident underscores the importance of strong patch management, proper segmentation, and comprehensive auditing to ensure that future intrusions are detected and contained before they spread.

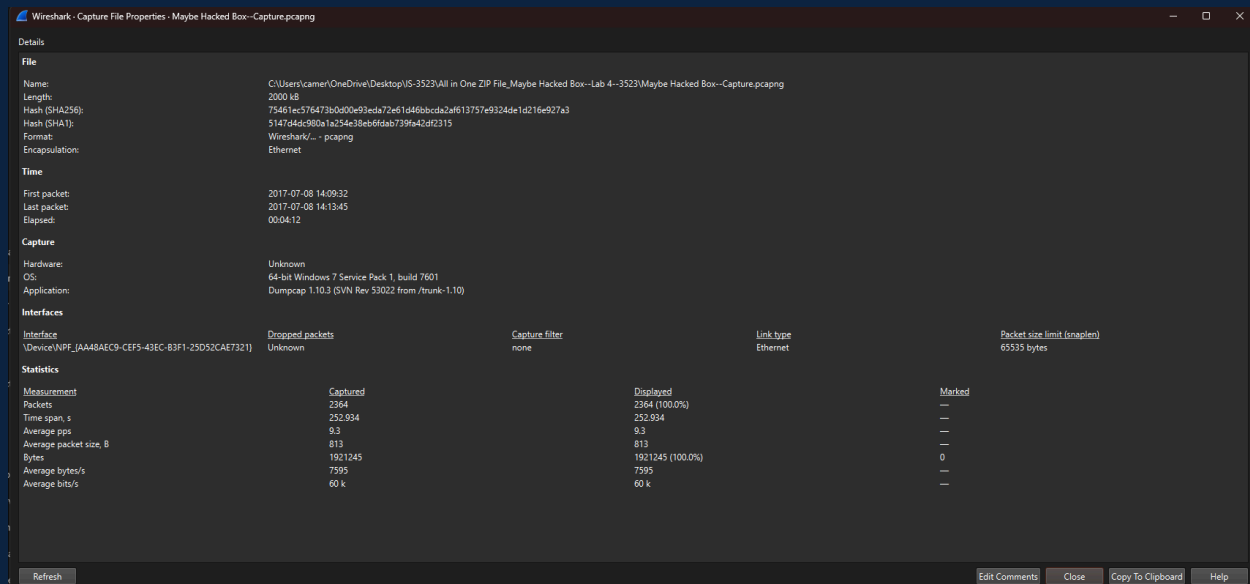
References

- BetaFred. n.d. “Microsoft Security Bulletin MS08-067 - Critical.” Microsoft Learn. <https://learn.microsoft.com/en-us/security-updates/securitybulletins/2008/ms08-067>.
- ———. n.d. “Microsoft Security Bulletin MS17-010 - Critical.” Microsoft Learn. <https://learn.microsoft.com/en-us/security-updates/securitybulletins/2017/ms17-010>.
- “What Is an SMB Exploit? How It Works & Examples | Twingate.” n.d. <https://www.twingate.com/blog/glossary/smb%20exploit>.
- “Windows Security Log Encyclopedia.” n.d. <https://www.ultimatewindowssecurity.com/securitylog/encyclopedia/default.aspx>.
- Winters, Kyle. 2025. “Stop Server Message Block (SMB) Exploits and Advance Your Career in Cybersecurity.” Cisco Blogs. May 22, 2025.

<https://blogs.cisco.com/learning/stop-smb-exploits-and-advance-your-career-in-cybersecurity>.

Appendix

- **Figure 1: Screenshot of the Capture.pcapng Properties**



- **Figure 2: Screenshot of the Capture.pcapng Protocol Hierarchy**

The screenshot shows the 'Wireshark - Protocol Hierarchy Statistics - Maybe Hacked Box--Capture.pcapng' window. It displays a table of protocol statistics, including the protocol name, percentage of packets, number of packets, percentage of bytes, number of bytes, bits per second, end packets, end bytes, end bits per second, and number of PDUs.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	2364	100.0	1921245	60 k	0	0	0	2364
Ethernet	100.0	2364	1.9	36092	1141	0	0	0	2364
Internet Protocol Version 6	2.5	59	0.1	2560	80	0	0	0	59
User Datagram Protocol	1.0	23	0.0	184	5	0	0	0	23
Link-local Multicast Name Resolution	0.7	16	0.0	528	16	16	528	16	16
DHCPv6	0.3	7	0.0	665	21	7	665	21	7
Internet Control Message Protocol v6	1.5	36	0.0	788	24	36	788	24	36
Internet Protocol Version 4	96.6	2284	2.4	45804	1448	0	0	0	2284
User Datagram Protocol	3.3	77	0.0	616	19	0	0	0	77
Teredo IPv6 over UDP tunneling	0.5	11	0.0	671	21	0	0	0	11
Simple Service Discovery Protocol	0.3	8	0.1	1392	44	8	1392	44	8
NetBIOS Name Service	0.6	15	0.1	1020	32	15	1020	32	15
NetBIOS Datagram Service	0.0	1	0.0	82	2	0	0	0	1
SMB (Server Message Block Protocol)	0.0	1	0.0	119	3	0	0	0	1
SMB MailSlot Protocol	0.0	1	0.0	25	0	0	0	0	1
Microsoft Windows Browser Protocol	0.0	1	0.0	33	1	1	33	1	1
Link-local Multicast Name Resolution	0.7	16	0.0	528	16	16	528	16	16
Domain Name System	0.8	20	0.1	1086	34	20	1086	34	20
Data	0.3	6	0.0	240	7	6	240	7	6
Transmission Control Protocol	90.4	2138	2.5	48292	1527	1648	36632	1158	2138
TCP Encapsulation of IPsec Packets	0.0	1	0.1	1460	46	0	0	0	1
Encapsulating Security Payload	0.0	1	0.1	1444	45	1	1444	45	1
SICK RKS12	0.1	3	5.5	105488	3336	1	220	6	5
Malformed Packet	0.1	2	0.0	0	0	1	0	0	2
NetBIOS Session Service	6.4	152	1.3	25415	803	0	0	0	152
SMB (Server Message Block Protocol)	6.4	152	1.3	24807	784	97	8222	260	152
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	1.5	35	0.7	13192	417	20	9620	304	35
Server Service	0.3	7	0.1	972	30	7	972	30	7
Microsoft Spool Subsystem	0.3	8	0.1	2240	70	6	2200	69	8
Malformed Packet	0.1	2	0.0	0	0	2	0	0	2
Data	0.8	20	0.3	4886	154	20	4886	154	20
Data	14.2	335	27.8	534441	16 k	335	480516	15 k	336
Internet Group Management Protocol	1.3	31	0.0	512	16	31	512	16	31
Internet Control Message Protocol	1.6	38	0.2	3152	99	38	3152	99	38
Address Resolution Protocol	1.4	32	0.0	896	28	32	896	28	32

● Figure 3: Screenshot of the DNS Filter

No.	Time	Source	Destination	Protocol	Length	Info
76	7.985202	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x6f4d A dns.msftncsl.com
79	7.988361	192.168.134.2	192.168.134.158	DNS	92	Standard query response 0x6f4d A dns.msftncsl.com A 131.107.255.255
80	7.988618	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x9207 AAAA dns.msftncsl.com
81	7.989342	192.168.134.2	192.168.134.158	DNS	104	Standard query response 0x9207 AAAA dns.msftncsl.com AAAA fd3e:4f5a:5b01::1
1651	69.895748	192.168.134.158	192.168.134.2	DNS	85	Standard query 0x85cb A teredo.ipv6.microsoft.com
1652	69.944857	192.168.134.2	192.168.134.158	DNS	182	Standard query response 0x85cb A teredo.ipv6.microsoft.com CNAME onpremiindows.ipv6.microsoft.com.akadns.net CNAME onpremy2.ipv6.microsoft.com.akadns.net A 65.55.158.118
1690	72.975801	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x8c8d A dns.msftncsl.com
1692	73.009451	192.168.134.2	192.168.134.158	DNS	92	Standard query response 0x8c8d A dns.msftncsl.com A 131.107.255.255
1693	73.009641	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x6f4d AAAA dns.msftncsl.com
1694	73.010179	192.168.134.2	192.168.134.158	DNS	104	Standard query response 0x6f4d AAAA dns.msftncsl.com AAAA fd3e:4f5a:5b01::1
1796	137.979759	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x8528 A dns.msftncsl.com
1797	138.029095	192.168.134.2	192.168.134.158	DNS	92	Standard query response 0x8528 A dns.msftncsl.com A 131.107.255.255
1798	138.029175	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x6f4d AAAA dns.msftncsl.com
1799	138.029942	192.168.134.2	192.168.134.158	DNS	104	Standard query response 0x6f4d AAAA dns.msftncsl.com AAAA fd3e:4f5a:5b01::1
2220	233.156205	192.168.134.158	192.168.134.2	DNS	85	Standard query 0x35dc A teredo.ipv6.microsoft.com
2221	233.157125	192.168.134.2	192.168.134.158	DNS	182	Standard query response 0x35dc A teredo.ipv6.microsoft.com CNAME onpremiindows.ipv6.microsoft.com.akadns.net CNAME onpremy2.ipv6.microsoft.com.akadns.net A 65.55.158.118
2338	237.976335	192.168.134.158	192.168.134.2	DNS	76	Standard query 0xe09c A dns.msftncsl.com
2339	238.021958	192.168.134.2	192.168.134.158	DNS	92	Standard query response 0xe09c A dns.msftncsl.com A 131.107.255.255
2340	238.022179	192.168.134.158	192.168.134.2	DNS	76	Standard query 0x076e AAAA dns.msftncsl.com
2341	238.022801	192.168.134.2	192.168.134.158	DNS	104	Standard query response 0x076e AAAA dns.msftncsl.com AAAA fd3e:4f5a:5b01::1

● Figure 4: Screenshot of the SMB Filter

No.	Time	Source	Destination	Protocol	Length	Info
105	8.083395	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400e, 417 bytes
115	8.093828	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400e, 334 bytes
107	8.085289	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400e, 315 bytes
113	8.091915	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400e, 239 bytes
117	8.095736	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400e, 135 bytes
2314	234.038581	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 732 bytes
97	8.069131	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 564 bytes
91	8.062835	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 460 bytes
2324	234.049156	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 302 bytes
2326	234.051181	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 262 bytes
93	8.064843	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400d, 228 bytes
75	7.951218	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 706 bytes
2308	234.027176	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 48 bytes
84	8.051044	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 48 bytes
2298	233.916103	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 364 bytes
73	7.949342	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 26 bytes
2304	234.022684	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 222 bytes
2301	234.019909	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400c, 14 bytes
2286	233.900798	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 697 bytes
2292	233.906656	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 52 bytes
67	7.938955	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 44 bytes
2288	233.902625	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 35 bytes
59	7.930423	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 243 bytes
63	7.934415	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 158 bytes
61	7.932482	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400b, 155 bytes
2278	233.880028	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 92 bytes
2272	233.881873	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 87 bytes
53	7.920052	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 68 bytes
47	7.913008	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 600 bytes
2270	233.879981	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 557 bytes
51	7.918198	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 16 bytes
104	8.083256	192.168.134.129	192.168.134.132	SMB	550	Write AndX Request, FID: 0x400e, 417 bytes at offset 197
104	8.093702	192.168.134.129	192.168.134.132	SMB	467	Write AndX Request, FID: 0x400e, 334 bytes at offset 853
112	8.091788	192.168.134.129	192.168.134.132	SMB	372	Write AndX Request, FID: 0x400e, 239 bytes at offset 923
90	8.062743	192.168.134.129	192.168.134.132	SMB	593	Write AndX Request, FID: 0x400d, 460 bytes at offset 502
2323	234.049029	192.168.134.129	192.168.134.132	SMB	435	Write AndX Request, FID: 0x400d, 302 bytes at offset 445
2302	233.032008	192.168.134.129	192.168.134.132	SMB	467	Write AndX Request, FID: 0x400c, 706 bytes at offset 370

- **Figure 5: Screenshot of the Frames Containing MZ**

Maybe Hacked Box--Capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

frame contains "MZ"

No.	Time	Source	Destination	Protocol	Length	Info
1548	39.905223	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1558151 Ack=109858 Win=65535 Len=1460
1515	39.904917	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1515811 Ack=109858 Win=65535 Len=1460
1490	39.904525	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1483691 Ack=109858 Win=65535 Len=1460
1448	39.903504	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1426751 Ack=109858 Win=65535 Len=1460
1417	35.784306	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1403403 Ack=109232 Win=65535 Len=1460
1415	35.784305	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1400483 Ack=109232 Win=65535 Len=1460
1386	35.783582	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1361063 Ack=109232 Win=65535 Len=1460
1363	35.783276	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1328943 Ack=109232 Win=65535 Len=1460
1236	35.781307	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1162503 Ack=109232 Win=65535 Len=1460
1215	35.780894	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1136223 Ack=109232 Win=65535 Len=1460
1199	35.780671	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=1115783 Ack=109232 Win=65535 Len=1460
1009	25.484189	192.168.134.132	192.168.134.129	TCP	1514	1241 → 29922 [ACK] Seq=15873 Ack=1043658 Win=64240 Len=1460
938	24.938850	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=969080 Ack=7039 Win=40880 Len=1460
759	12.933483	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=801777 Ack=399 Win=15544 Len=1460
728	12.932348	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=762357 Ack=399 Win=15544 Len=1460
724	12.932346	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=756517 Ack=399 Win=15544 Len=1460
485	8.120553	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=463309 Ack=1 Win=14600 Len=1460
128	8.115140	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=4385 Ack=1 Win=14600 Len=1460
125	8.115137	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=5 Ack=1 Win=14600 Len=1460
46	7.912909	192.168.134.129	192.168.134.132	DCERPC	733	Bind: call_id: 0, Fragment: Single, 13 context items: 0e042bc0-cab3-517d-523f-7cbe71a19dd5 V5.1 (3...

- **Figure 6: Screenshot of Event Viewer**

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
- Applications and Services Logs
- Saved Logs
 - Maybe Hacked Box--Application Log
 - Maybe Hacked Box--Security Log
 - Subscriptions

Maybe Hacked Box--Security Log Page 1

Next Page Back to Top

To make this Analytic, Debug or Classic event log easier to navigate and manipulate, first save it in .evtx

Level	Date and Time	Source	Event ID	Task Ca...
Information	8/6/2016 10:00:32 AM	EventLog	6009	None
Information	8/6/2016 10:00:32 AM	EventLog	6005	None
Information	8/6/2016 10:00:26 AM	Serial	2	None
Information	8/6/2016 3:05:14 PM	EventLog	6011	None
Information	8/6/2016 3:06:04 PM	Workst...	3260	None
Information	8/6/2016 3:08:17 PM	HTTP	15007	None
Information	8/6/2016 3:10:55 PM	Setup	60054	None
Information	8/6/2016 3:11:24 PM	EventLog	6009	None
Information	8/6/2016 3:11:24 PM	EventLog	6005	None
Information	8/6/2016 3:11:43 PM	SRServi...	115	None
Information	8/6/2016 3:11:52 PM	Service ...	7035	None
Information	8/6/2016 3:11:52 PM	Service ...	7035	None
Information	8/6/2016 3:11:52 PM	Service ...	7035	None
Information	8/6/2016 3:11:52 PM	Service ...	7036	None
Information	8/6/2016 3:11:52 PM	Service ...	7035	None
Information	8/6/2016 3:11:52 PM	Service ...	7036	None
Information	8/6/2016 3:11:52 PM	Service ...	7036	None
Information	8/6/2016 3:11:54 PM	Service ...	7036	None

General Details

Log Name:

Source: Logged:

Event ID: Task Category:

Level: Keywords:

User: Computer:

OpCode:

Actions

- Maybe Hacked Box--Security Log
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- View
- Delete
- Rename
- Refresh
- Help

- **Figure 7: Screenshot of the Network Endpoints**

Wireshark - Endpoints - Maybe Hacked Box--Capture.pcapng

Endpoint Settings

Name resolution
Limit to display filter

Copy
Map

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
65.55.158.118	11	1 kB	0	0 bytes	11	1 kB						
192.168.134.1	15	2 kB	15	2 kB	0	0 bytes						
192.168.134.2	53	6 kB	28	4 kB	25	2 kB						
192.168.134.129	2,158	2 MB	1,573	2 MB	585	166 kB						
192.168.134.132	2,173	2 MB	591	167 kB	1,582	2 MB						
192.168.134.158	96	6 kB	77	6 kB	19	2 kB						
192.168.134.255	7	735 bytes	0	0 bytes	7	735 bytes						
224.0.0.22	31	2 kB	0	0 bytes	31	2 kB						
224.0.0.252	16	1 kB	0	0 bytes	16	1 kB						
239.255.255.250	8	2 kB	0	0 bytes	8	2 kB						

- **Figure 8: Screenshot of the SYN flag Filter**

Maybe Hacked Box--Capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
28	7.871742	192.168.134.129	192.168.134.132	TCP	74	41254 → 445 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1143380 TSecr=0 WS=128
121	8.097467	192.168.134.132	192.168.134.129	TCP	62	1241 → 29922 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
1701	76.950586	192.168.134.129	192.168.134.132	TCP	74	51865 → 1000 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1160650 TSecr=0 WS=128
1897	230.025957	192.168.134.129	192.168.134.132	TCP	74	53768 → 69 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1899	230.026334	192.168.134.129	192.168.134.132	TCP	74	42832 → 80 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1901	230.026698	192.168.134.129	192.168.134.132	TCP	74	54297 → 49 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1903	230.027067	192.168.134.129	192.168.134.132	TCP	74	44382 → 81 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1905	230.027441	192.168.134.129	192.168.134.132	TCP	74	43410 → 105 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1907	230.027784	192.168.134.129	192.168.134.132	TCP	74	42350 → 42 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1909	230.028215	192.168.134.129	192.168.134.132	TCP	74	52129 → 25 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198919 TSecr=0 WS=128
1911	230.029992	192.168.134.129	192.168.134.132	TCP	74	37992 → 23 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198920 TSecr=0 WS=128
1913	230.031501	192.168.134.129	192.168.134.132	TCP	74	48859 → 22 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198920 TSecr=0 WS=128
1915	230.031976	192.168.134.129	192.168.134.132	TCP	74	37485 → 21 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198920 TSecr=0 WS=128
1917	230.175354	192.168.134.129	192.168.134.132	TCP	74	40977 → 143 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198956 TSecr=0 WS=128
1919	230.175693	192.168.134.129	192.168.134.132	TCP	74	50284 → 161 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198956 TSecr=0 WS=128
1921	230.176041	192.168.134.129	192.168.134.132	TCP	74	50414 → 389 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198956 TSecr=0 WS=128
1923	230.176416	192.168.134.129	192.168.134.132	TCP	74	42067 → 407 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198956 TSecr=0 WS=128
1925	230.177161	192.168.134.129	192.168.134.132	TCP	74	53080 → 139 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1927	230.177481	192.168.134.129	192.168.134.132	TCP	74	56190 → 135 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1931	230.177989	192.168.134.129	192.168.134.132	TCP	74	45115 → 123 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1933	230.178481	192.168.134.129	192.168.134.132	TCP	74	54760 → 113 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1942	230.180171	192.168.134.129	192.168.134.132	TCP	74	40743 → 111 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1944	230.180599	192.168.134.129	192.168.134.132	TCP	74	47601 → 110 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198957 TSecr=0 WS=128
1946	230.237414	192.168.134.129	192.168.134.132	TCP	74	33955 → 548 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1948	230.237684	192.168.134.129	192.168.134.132	TCP	74	40132 → 617 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1950	230.238016	192.168.134.129	192.168.134.132	TCP	74	41958 → 623 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1952	230.238370	192.168.134.129	192.168.134.132	TCP	74	35074 → 689 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1954	230.239633	192.168.134.129	192.168.134.132	TCP	74	43347 → 535 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1956	230.239945	192.168.134.129	192.168.134.132	TCP	74	34516 → 514 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1958	230.240403	192.168.134.129	192.168.134.132	TCP	74	45771 → 513 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1960	230.240798	192.168.134.129	192.168.134.132	TCP	74	55428 → 512 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198972 TSecr=0 WS=128
1962	230.242193	192.168.134.129	192.168.134.132	TCP	74	50952 → 445 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198973 TSecr=0 WS=128
1964	230.242604	192.168.134.129	192.168.134.132	TCP	74	33888 → 443 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198973 TSecr=0 WS=128
1970	230.295572	192.168.134.129	192.168.134.132	TCP	74	44464 → 1000 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198986 TSecr=0 WS=128
1972	230.295935	192.168.134.129	192.168.134.132	TCP	74	41479 → 1099 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198986 TSecr=0 WS=128
1974	230.296332	192.168.134.129	192.168.134.132	TCP	74	50795 → 1100 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198986 TSecr=0 WS=128
1976	230.296555	192.168.134.129	192.168.134.132	TCP	74	43373 → 1133 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM TSval=1198986 TSecr=0 WS=128